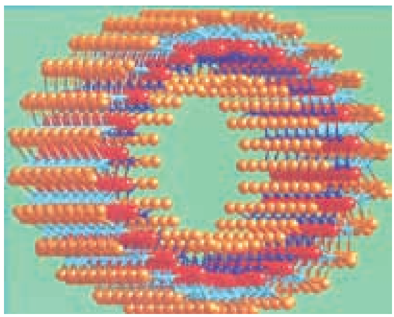
nanotubes, a measurement of how much force it takes to bend a material, is about 5 times higher than for steel. In addition to being strong and elastic, carbon nanotubes are also lightweight, with a density about one quarter that of steel.

As if that weren't enough, carbon nanotubes also conduct heat and cold really well (they have a high thermal conductivity); some researchers predict a thermal conductivity more than 10 times that of silver — and if you've ever picked up a fork from a hot stove, you know silver and other metals are pretty darn good conductors of heat.

CNTs are an example of the true power of nanotechnology. They open an incredible range of applications in materials science, electronics, chemical processing, energy management, and many other fields. It is already being used as Field Emitters, Energy Storage, Catalyst support, Biomedical applications among others.

Inorganic Nanotubes

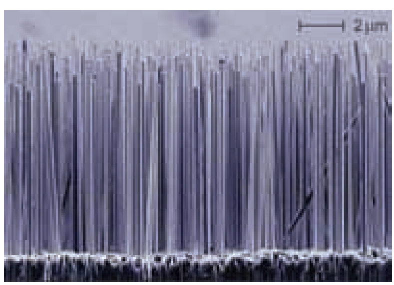
Inorganic nanotubes are based on layered compounds such as molybdenum disulphide and were discovered shortly after CNTs. They have excellent tribological (lubricating) properties, resistance to shockwave impact, catalytic reactivity, and high capacity for hydrogen and lithium storage, which suggest a range of promising applications.



Inorganic Nanotubes

Nanowires

Nanowires are ultrafine wires or linear arrays of dots, formed by self-assembly. They can be made from a wide range of materials. Semiconductor nanowires made of silicon, gallium nitride and indium phosphide has demonstrated remarkable optical, electronic and magnetic characteristics. Nanowires have potential



Silicon nanowires